

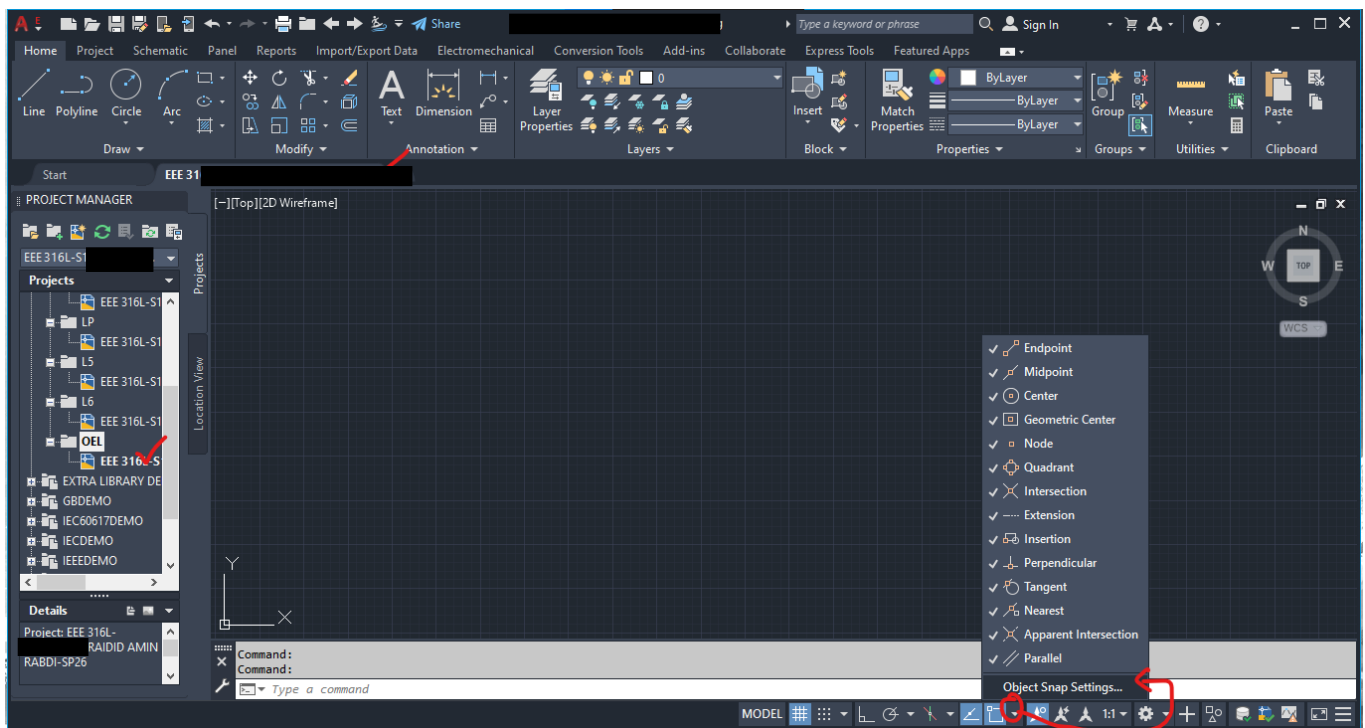
Problem

Suppose you are an Electrical Design Engineer at a reputed engineering firm. The firm has got a project to setup an emergency Power Station for the extension of Metro Rail Line 9, and you will be responsible for designing and drawing Single Line Diagram of an emergency Power Station in AutoCAD Electrical.

Note: The Metro Rail runs on 1100 Volts. So, make sure 1100 volts are fed into the Metro rail feeder to run the trains.

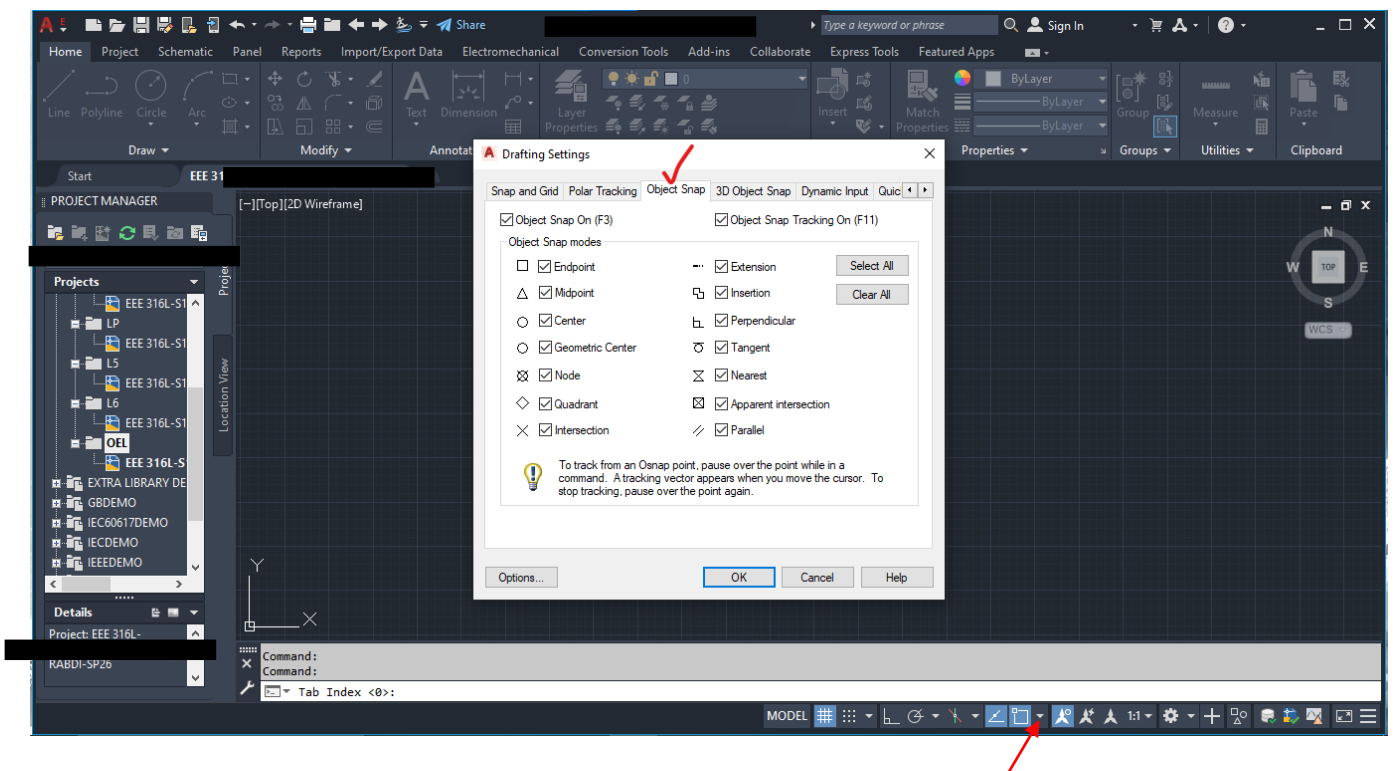
STEP 01:

Our drawing template is now successfully created and ready to draw with few parameters adjustment.



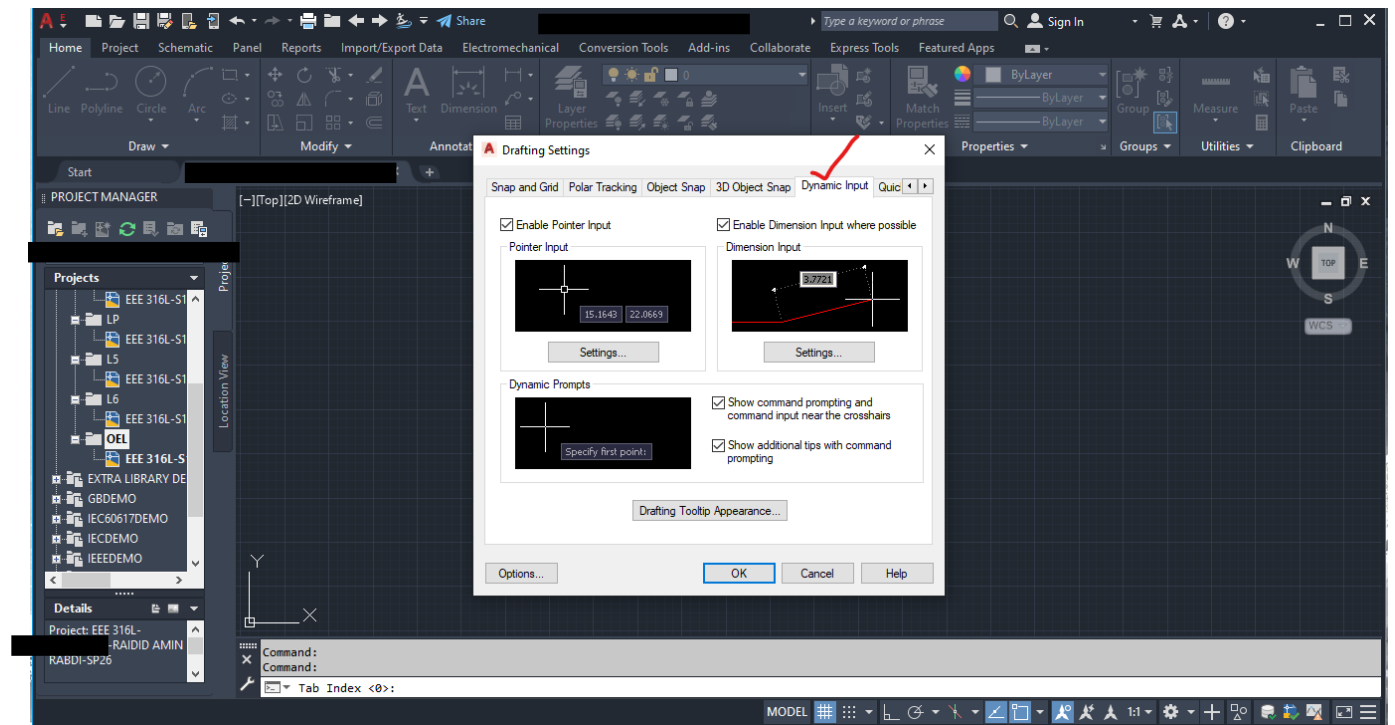
STEP 02:

Go to “OBJECT SNAP” and turn on all of these option. It can be found at “Object Snap Settings” indicated with an arrow.



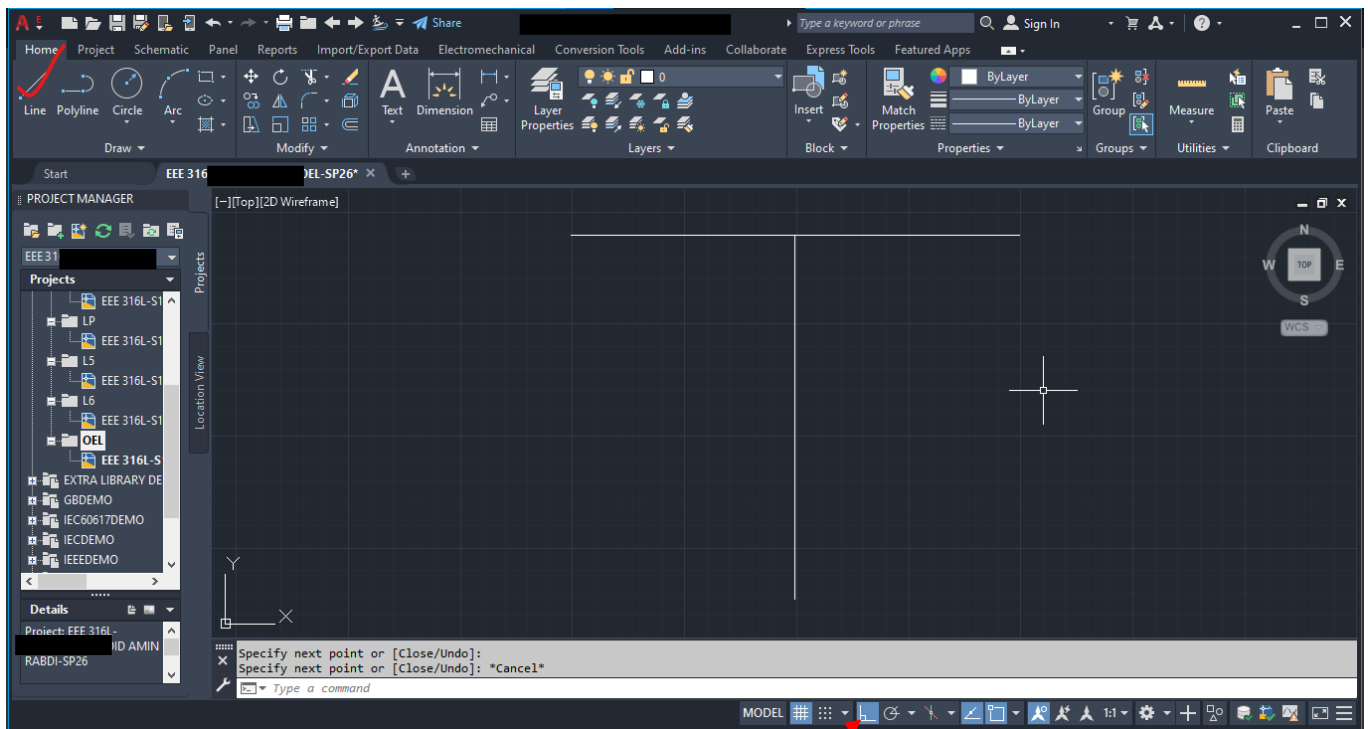
STEP 03:

Go to “DYNAMIC INPUT” and turn on all of these option. Click OK to start drawing.

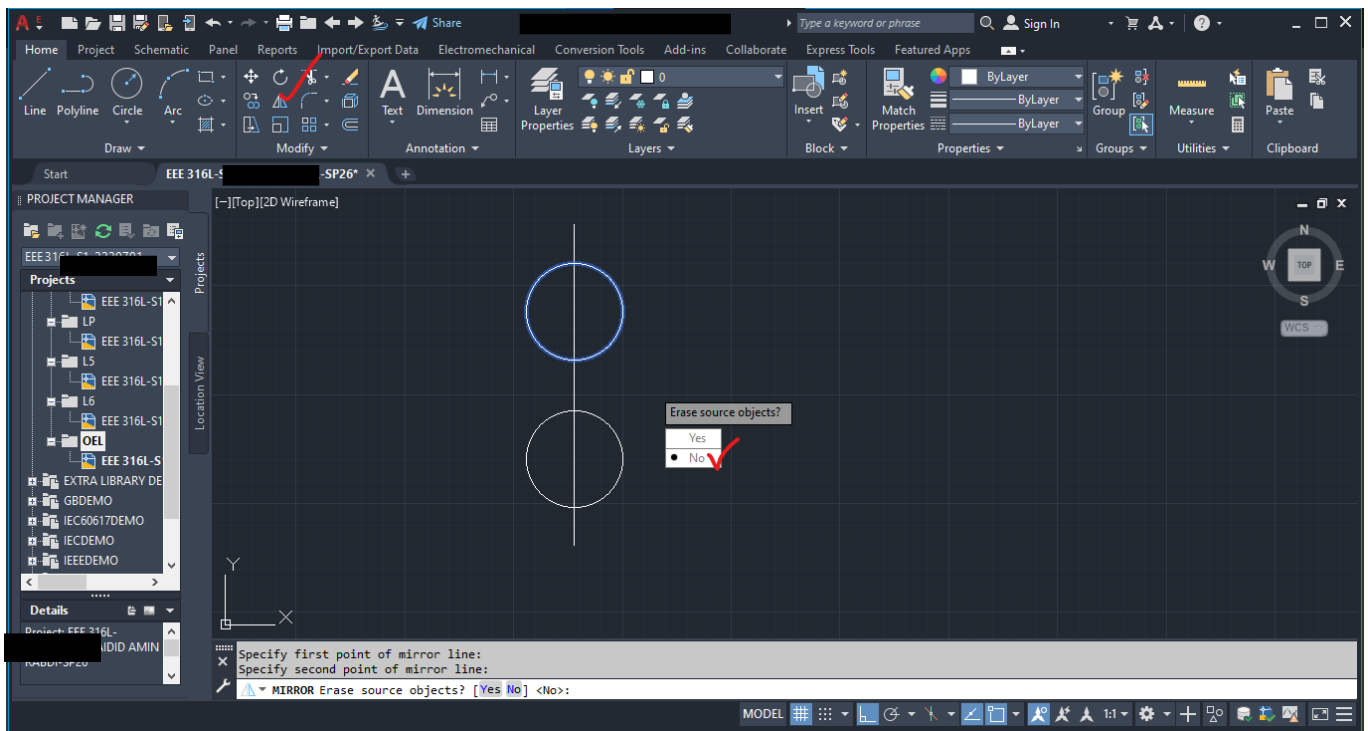


All the settings are now adjusted. Now I move on drawing my design →

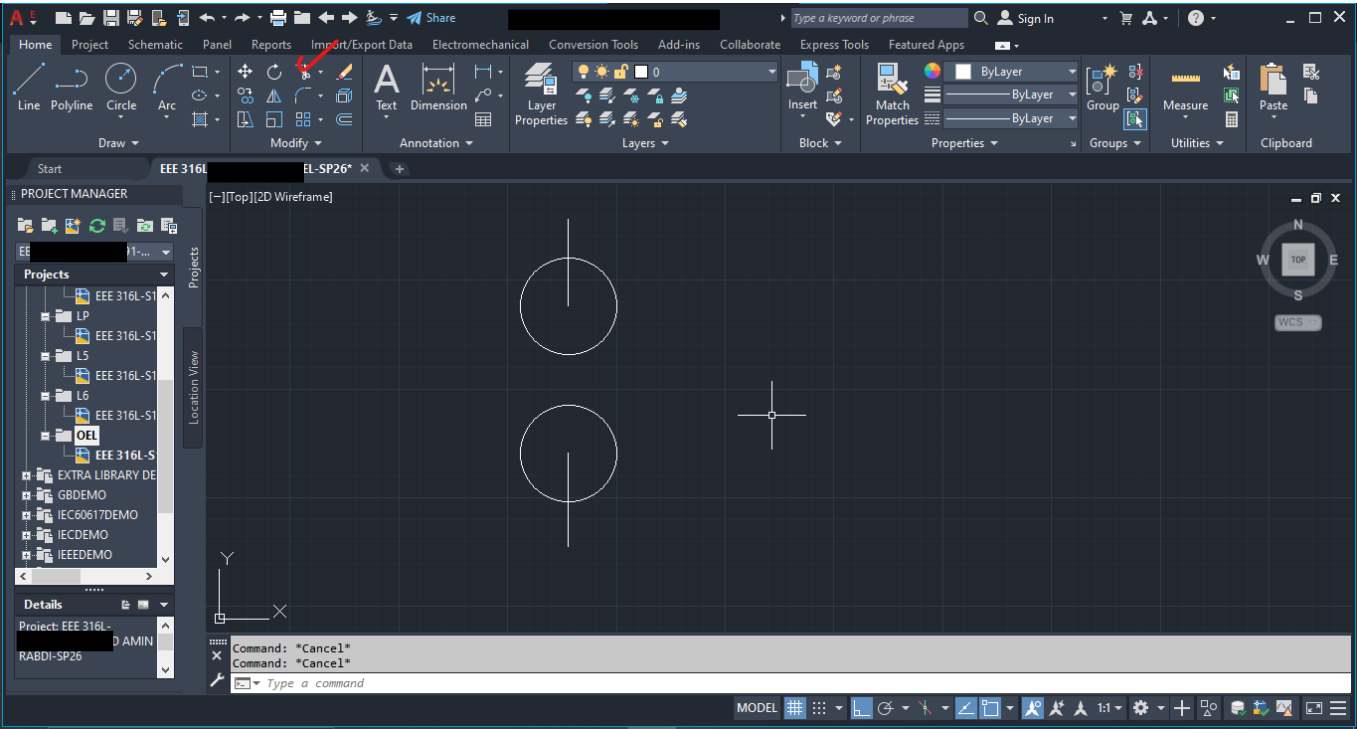
Starting with an 'T' shaped line using line tool. "ORTHO" is ON for straight line, setting icon at the bottom showed with an arrow.



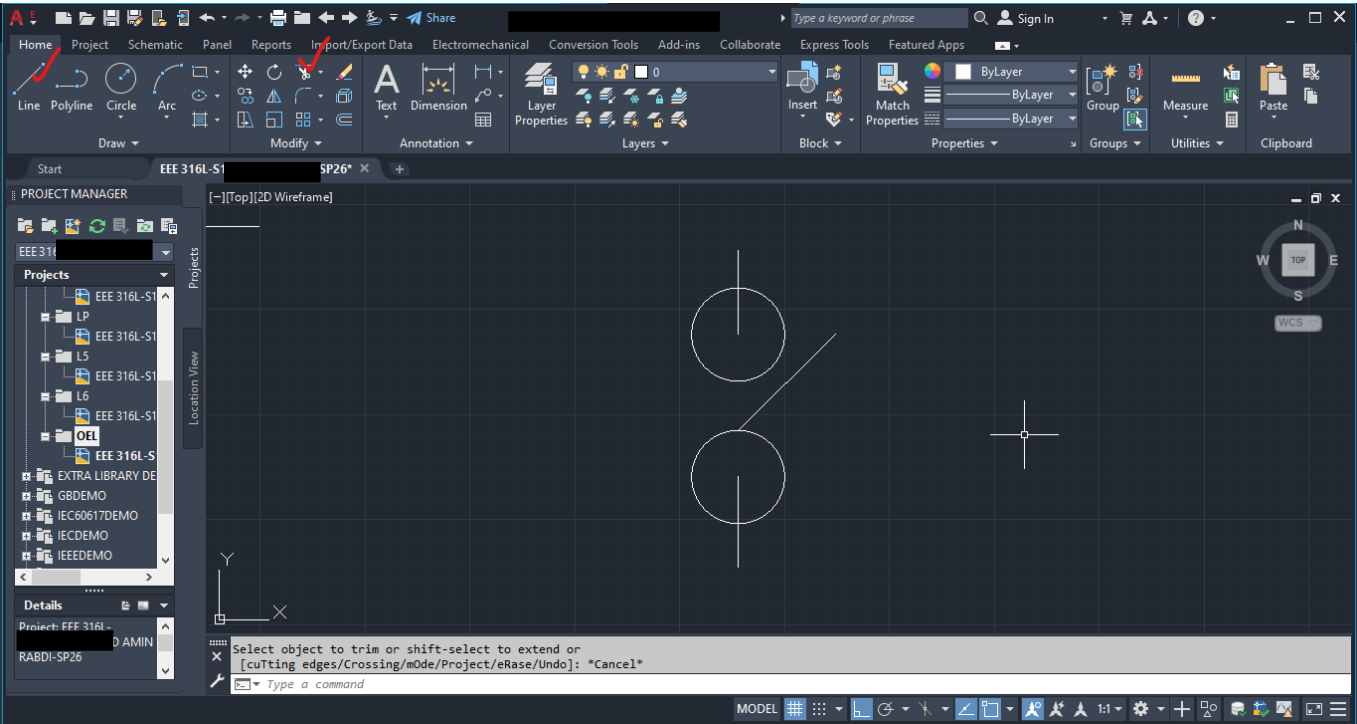
Select a circle on top of a line, MIRROR from the tools section used where source object is not erased.



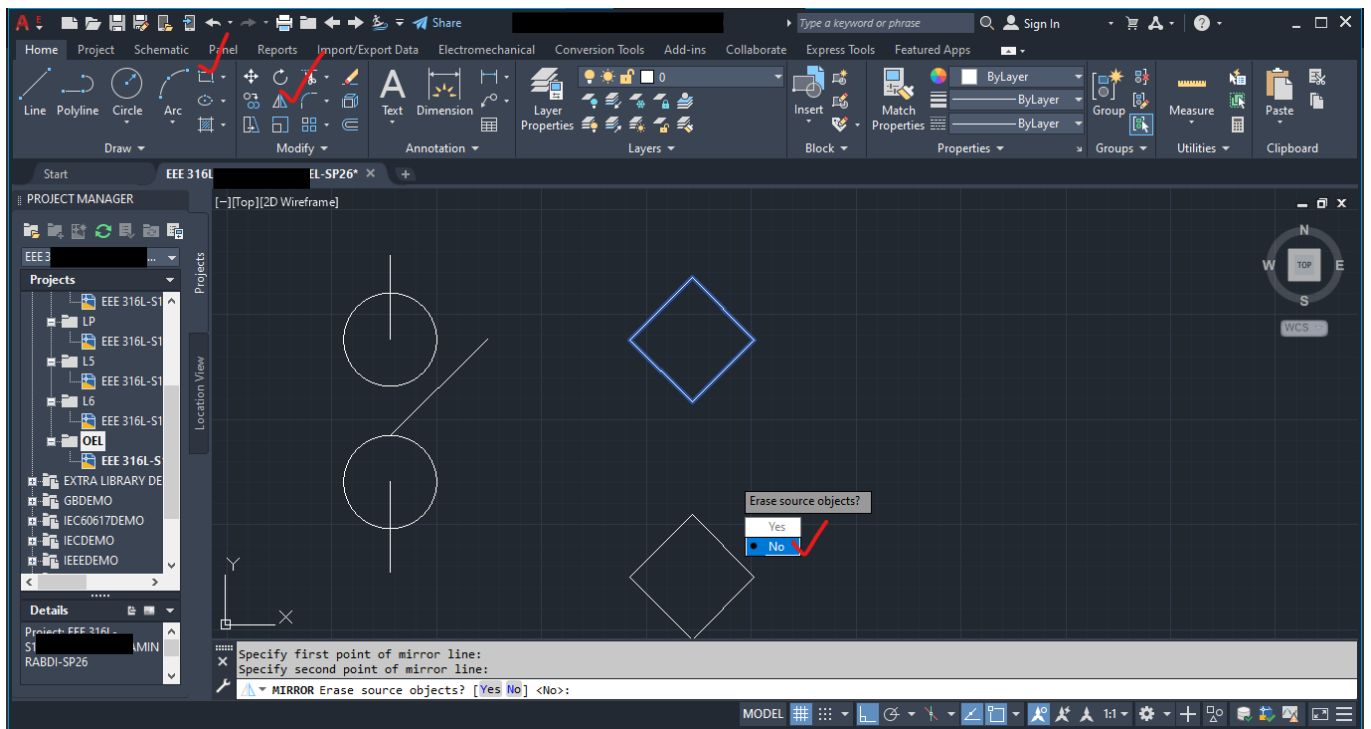
Trim unnecessary portions.



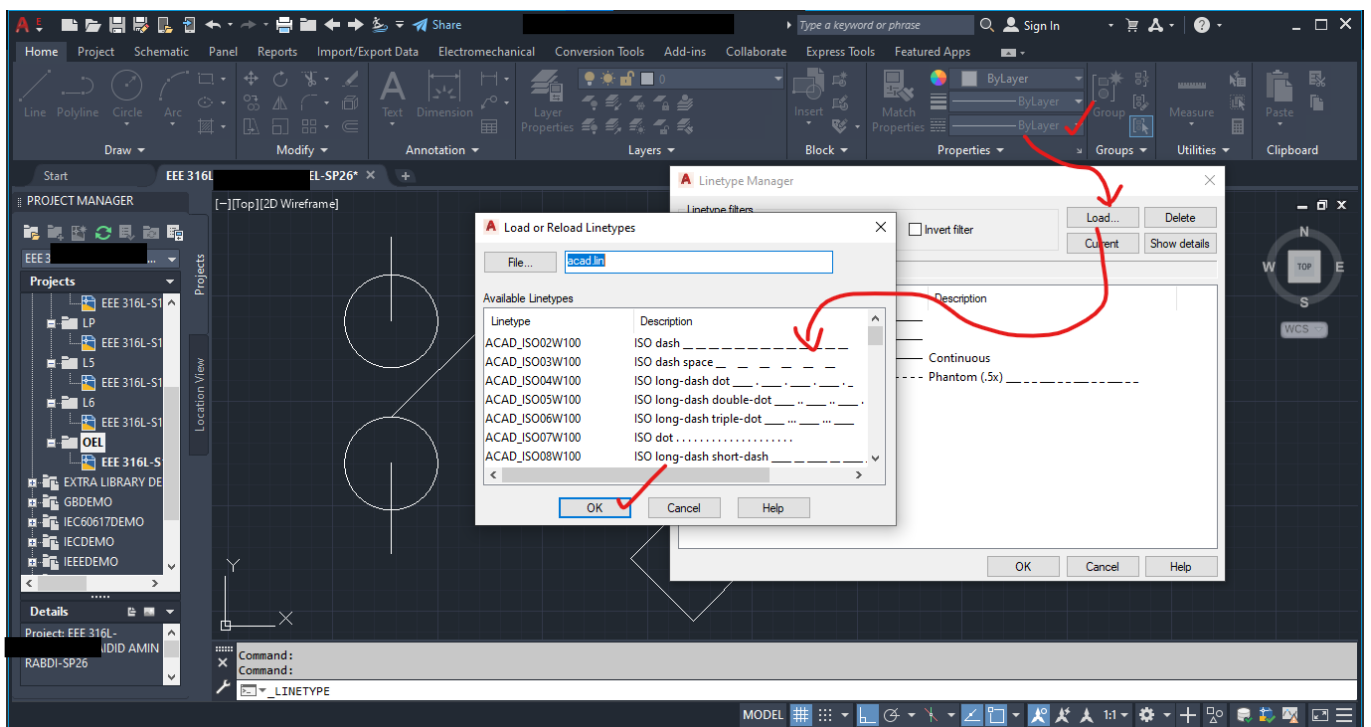
Add a line on top point of the lower circle at an angle. Thus, G.O. Switch designed.



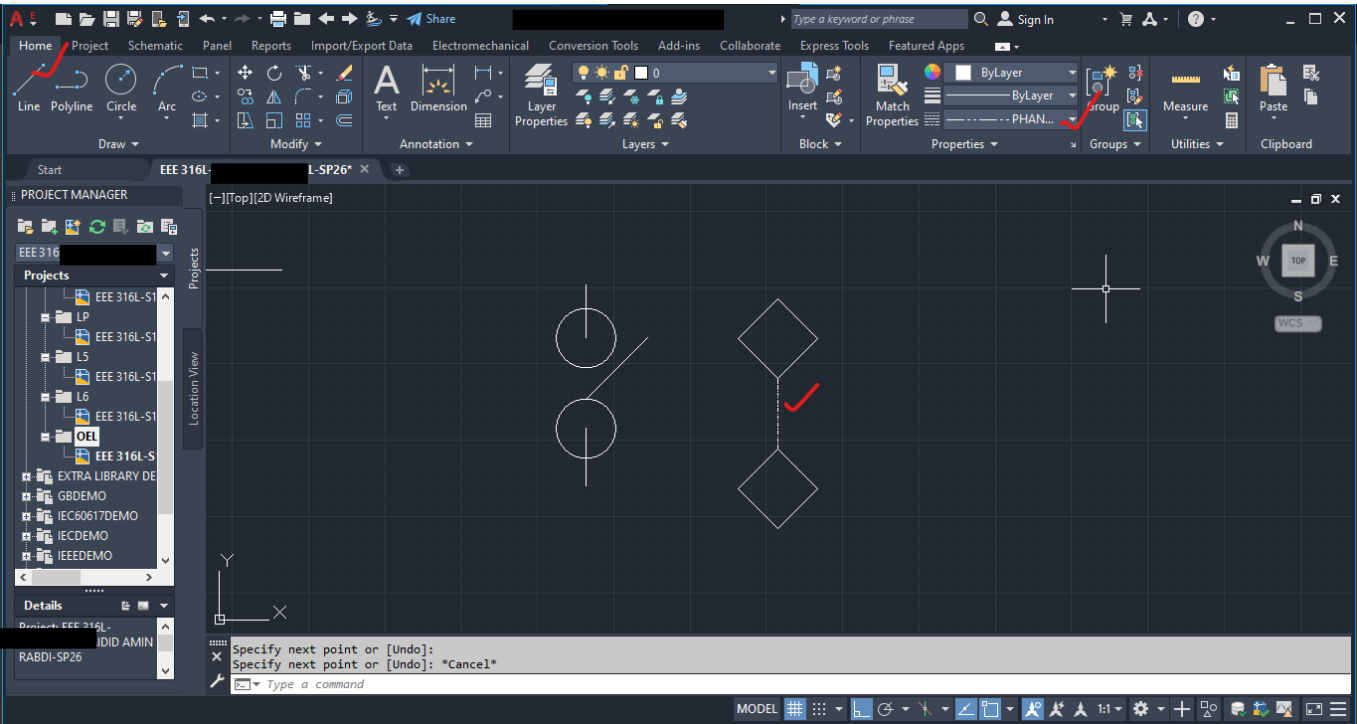
Cable Box: Select a rectangle using the square box. Use rotate to angle the box. Then copy and pasted at a distance.



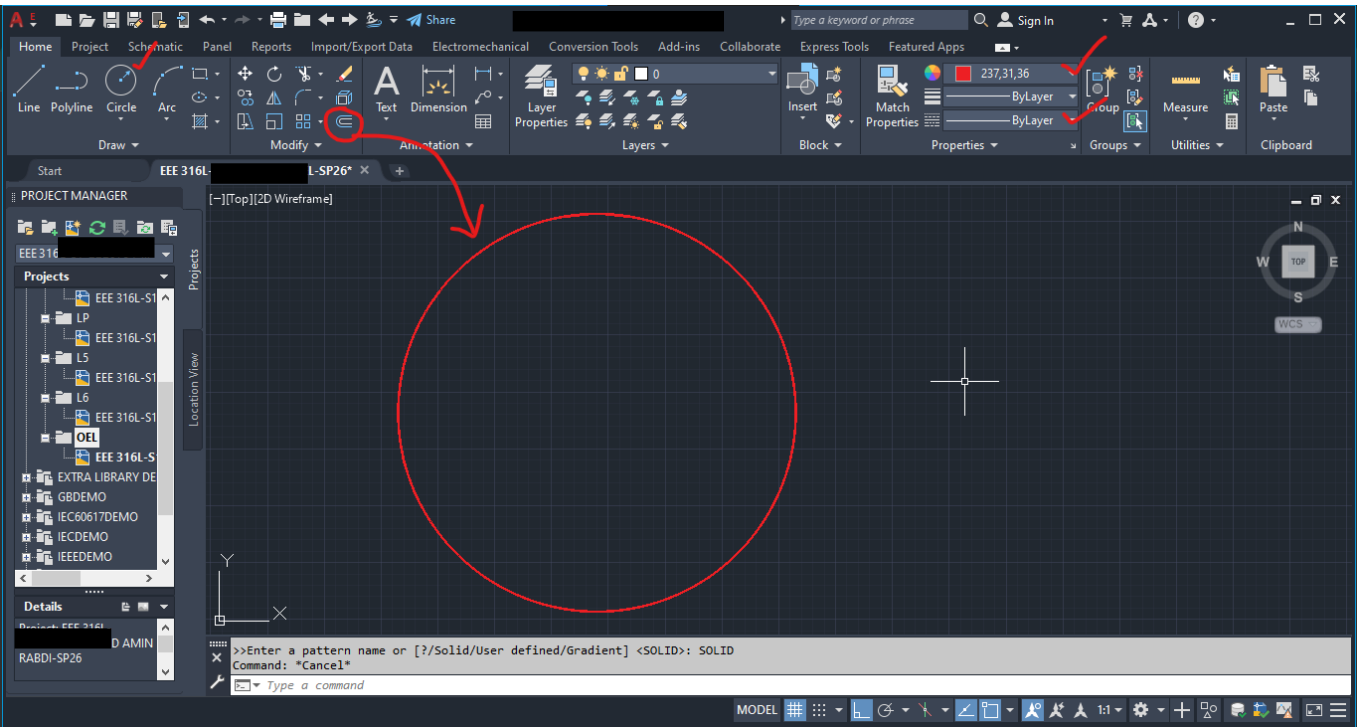
Underground Cable: Selected an appropriate layer to draw at my choice. Go to the BY LAYER → LOAD → any dotted line → OK



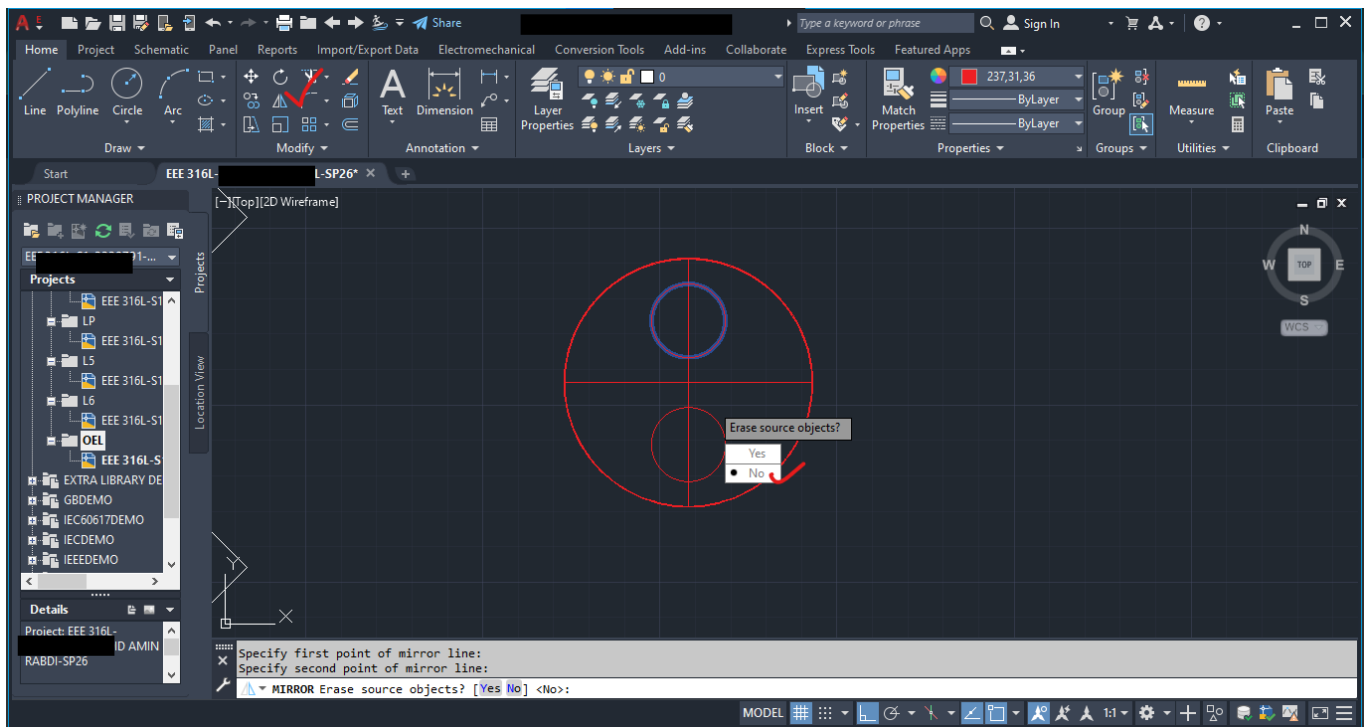
Cable Box and Underground Cable designed.



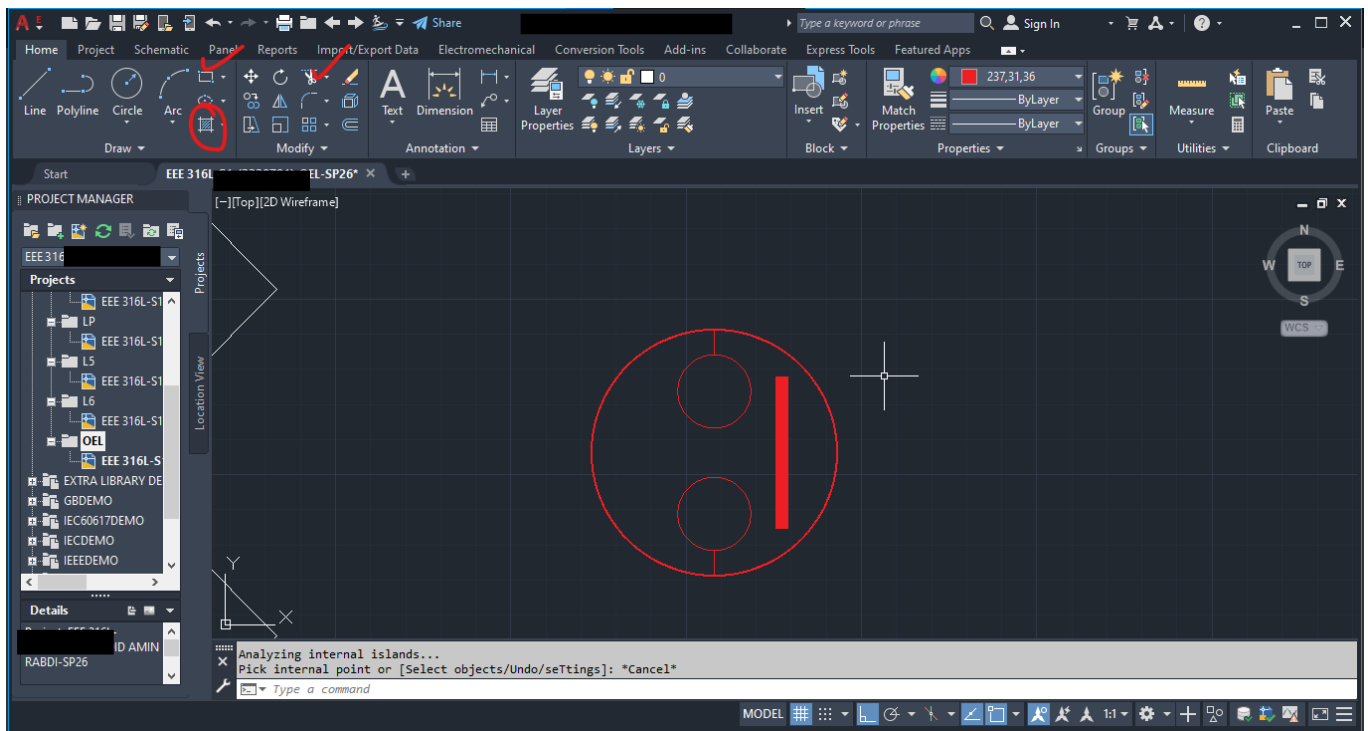
Change the ByLayer color from white to red, draw a circle. OFFSET tool used to bold the circle. Offset value is or 1” used.



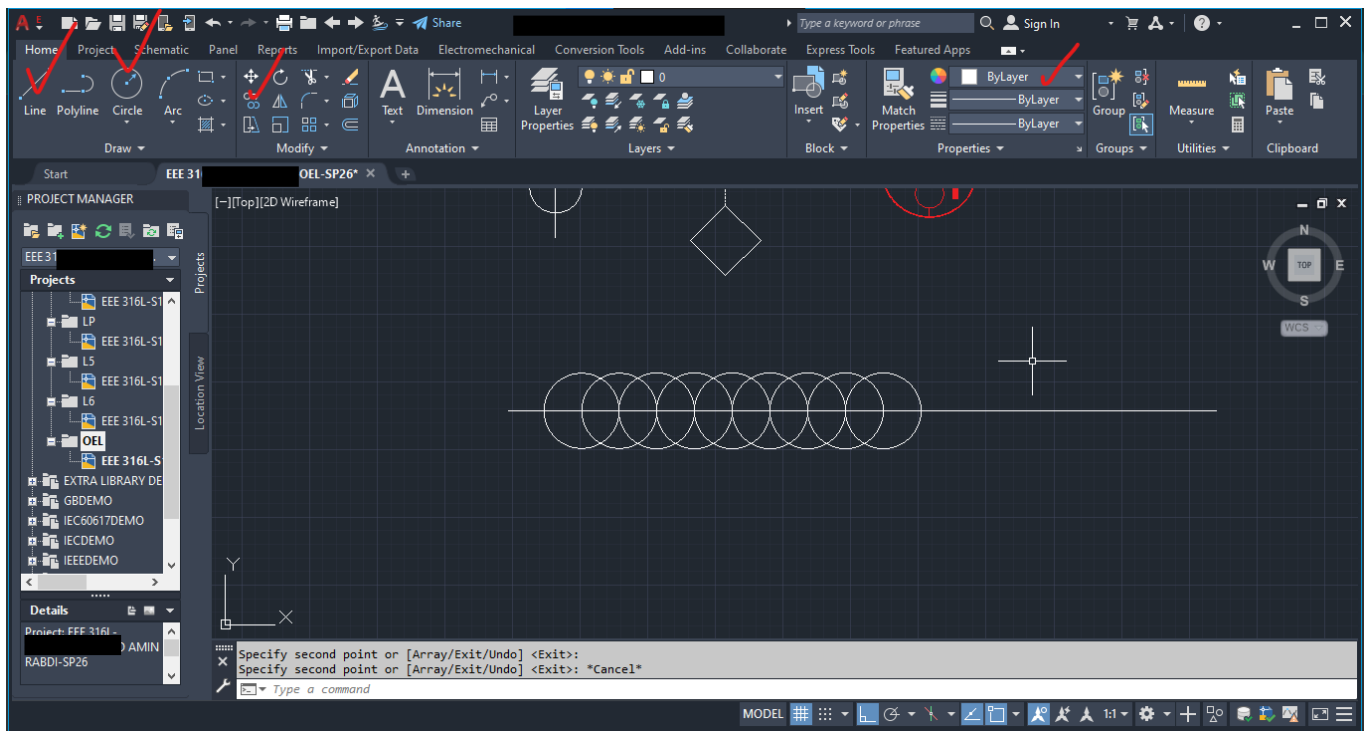
A big and A small circle. A line at the middle point added. Copy Pasted the small circle on top of the line. Then horizontal small lines touching the small circles to move the trimmed line.



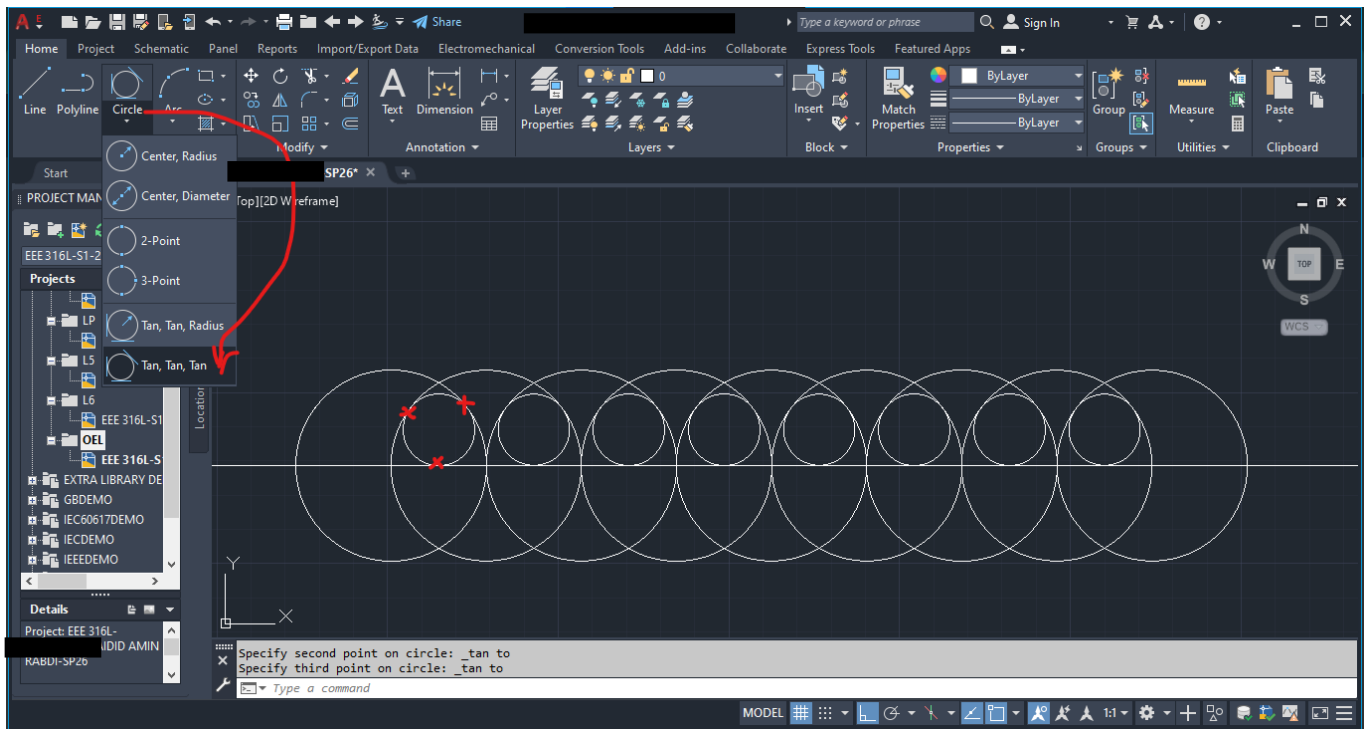
Trimmed unnecessary portions and for the bold portion, rectangle tool and HATCH tool used.



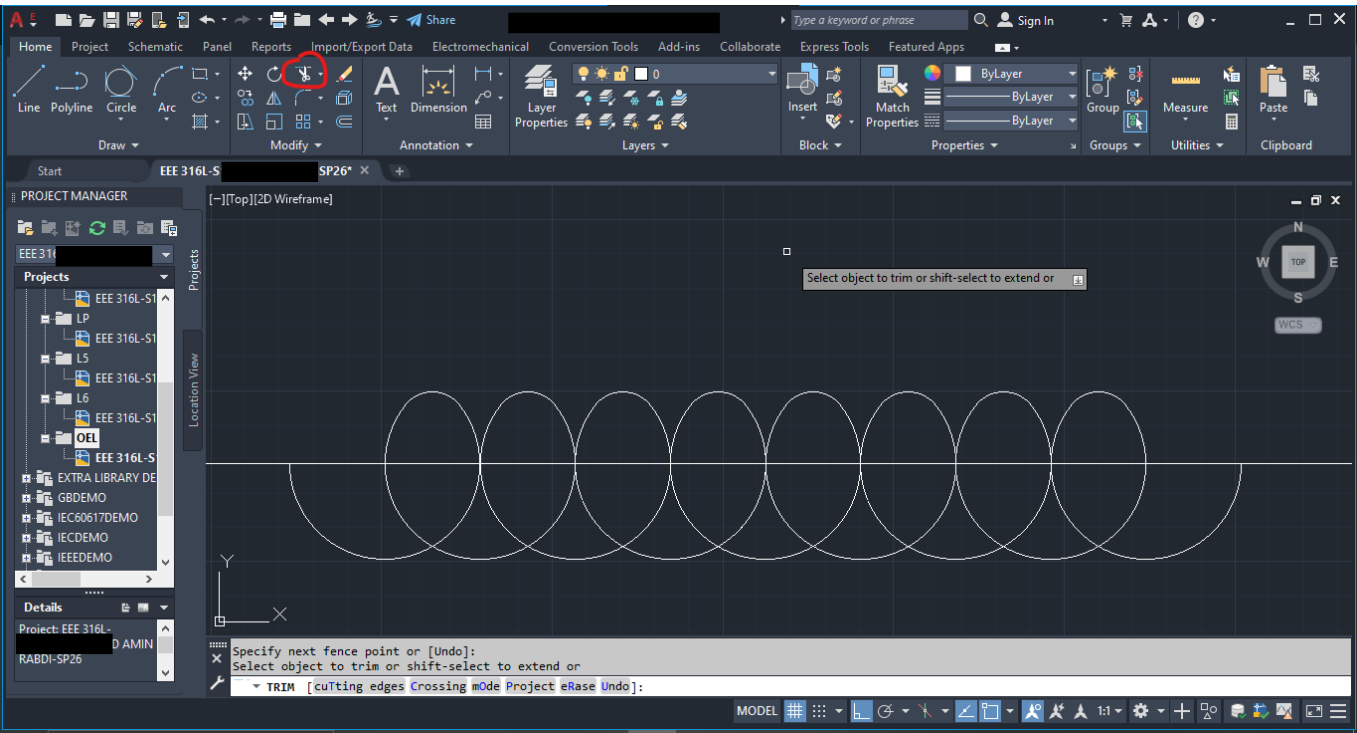
One Line and One circle. Copy and pasted the circle 8 times. Select the middle point of the circle after selecting the copy-paste tool and paste it on the right side middle point of the circle so that they intersect both center.



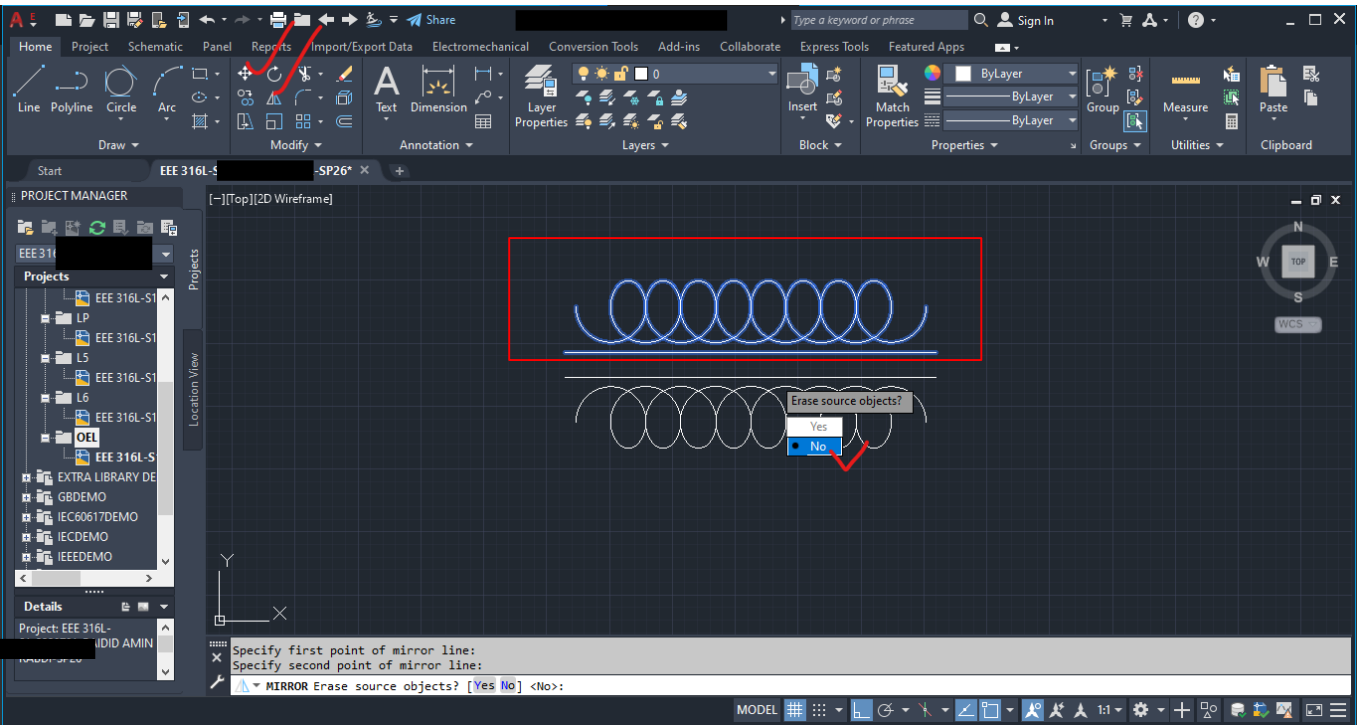
Tan, Tan, Tan selected, click 3 cross marked area and perfect circle inside will be automatically added.



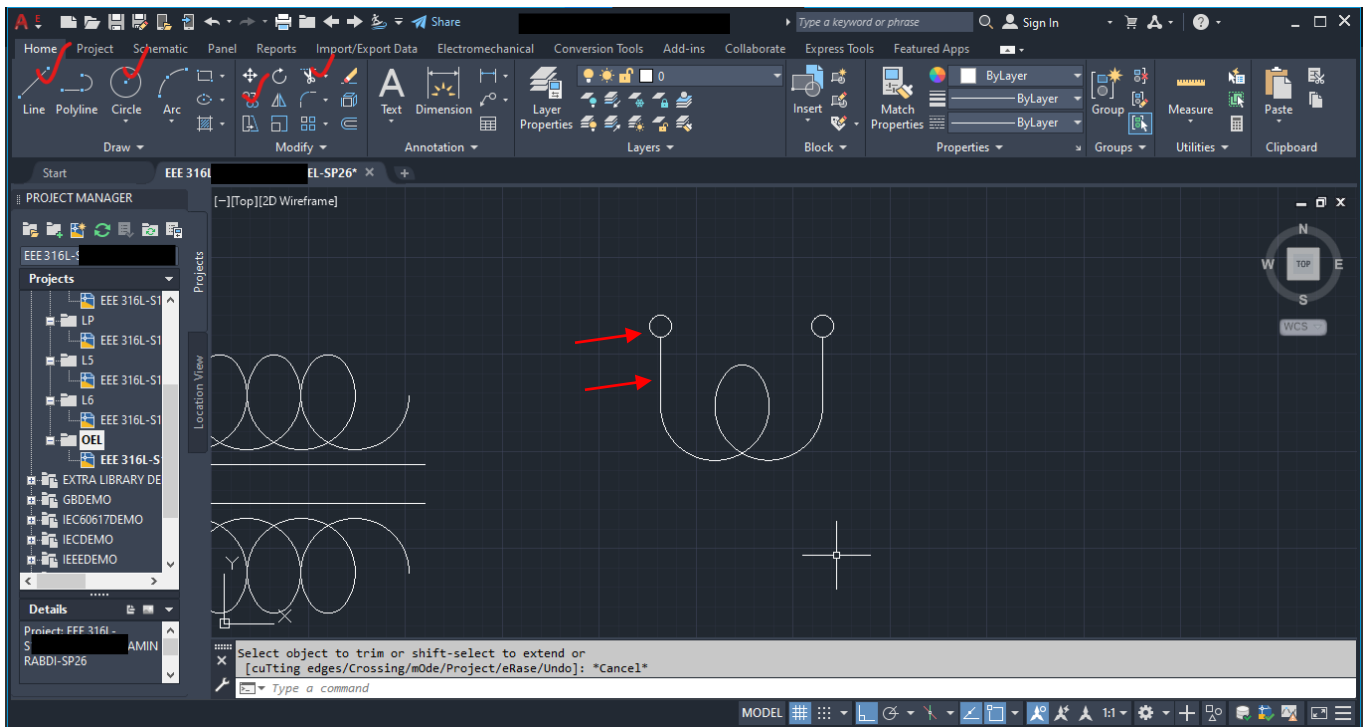
Trimmed unnecessary above portions.



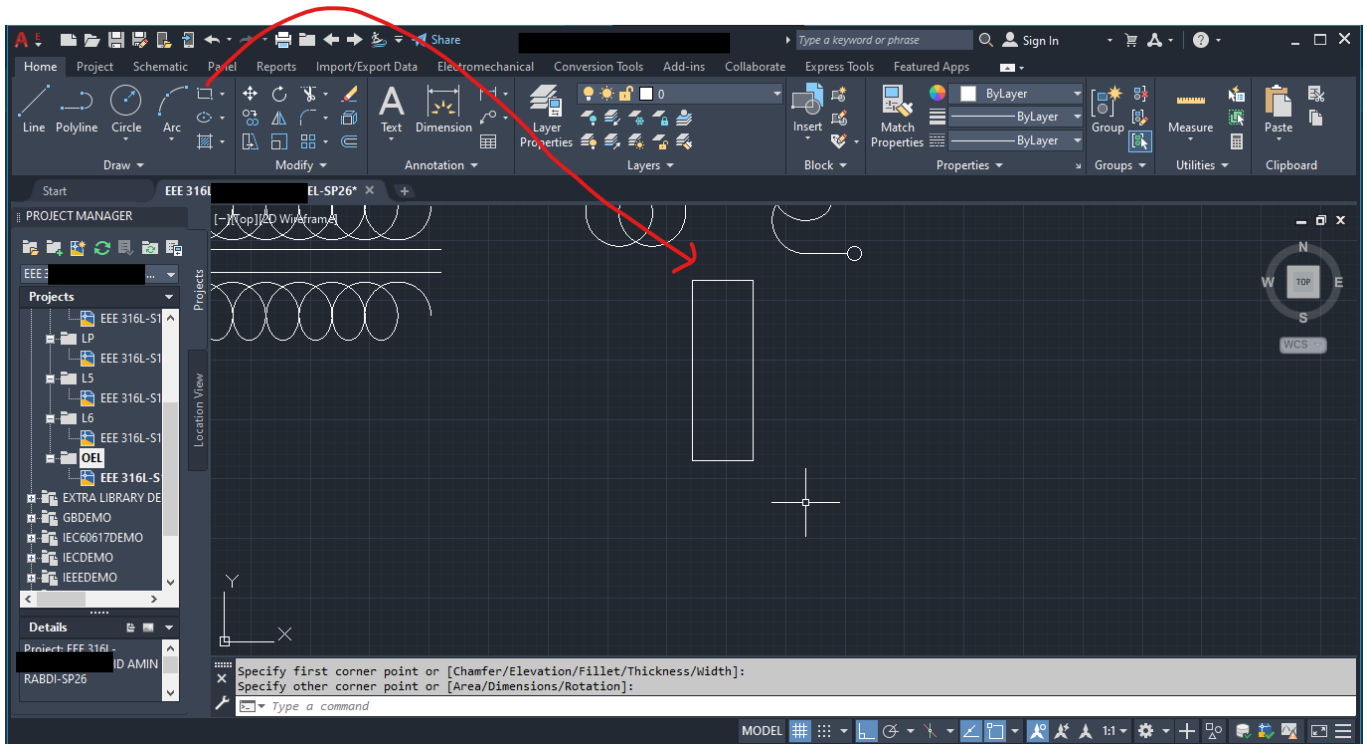
MIRROR tool used. But MOVE the Line first. Select the blue marked portion, mirror it selected slightly under the line to have the space.



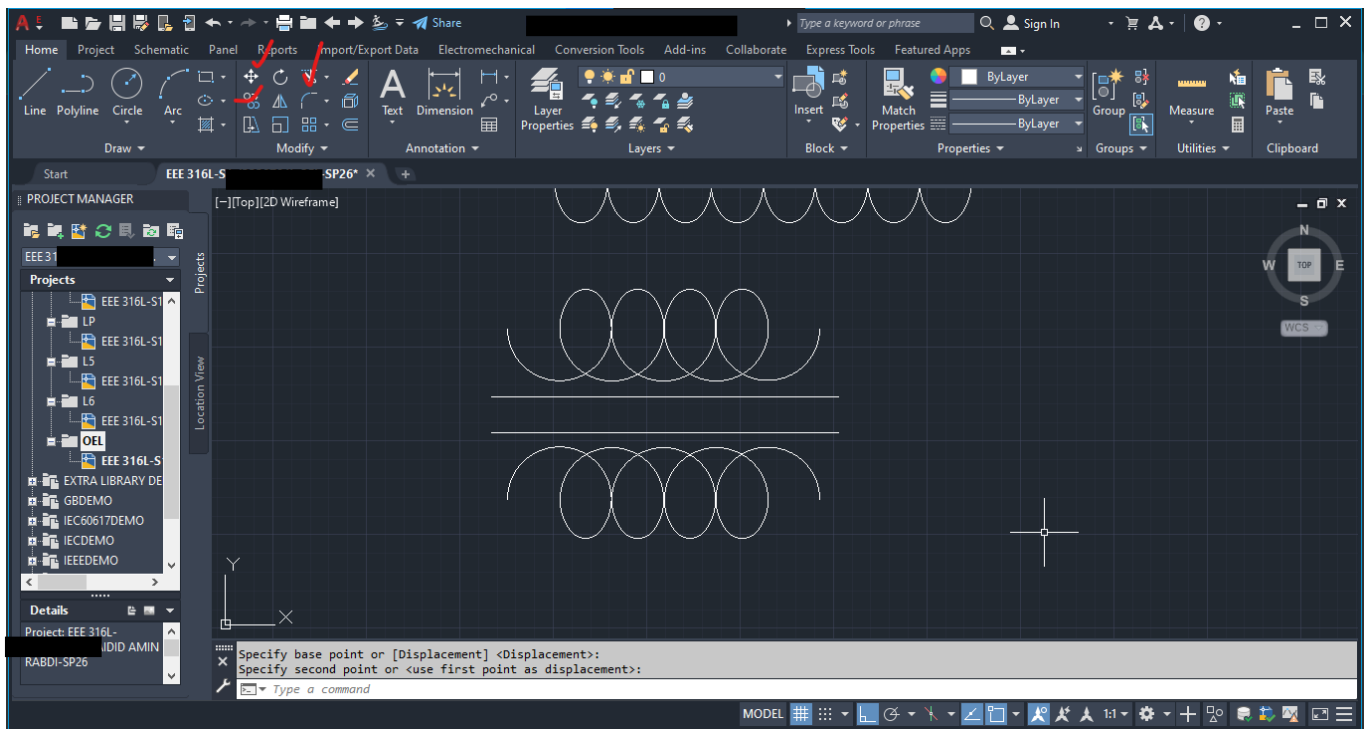
The coil copied from the transformer design. A line and small circle added and copied to have that on the other side also.



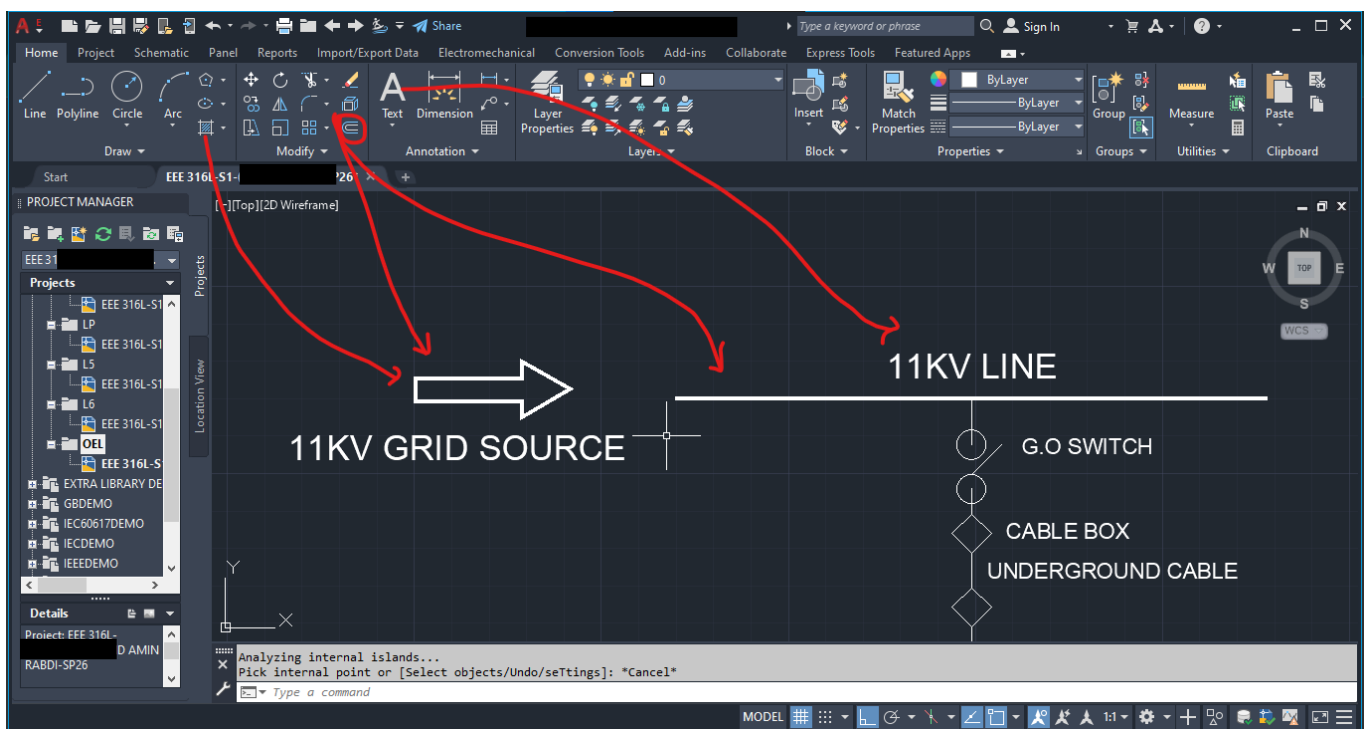
Use of Rectangle tool to draw fuse box.



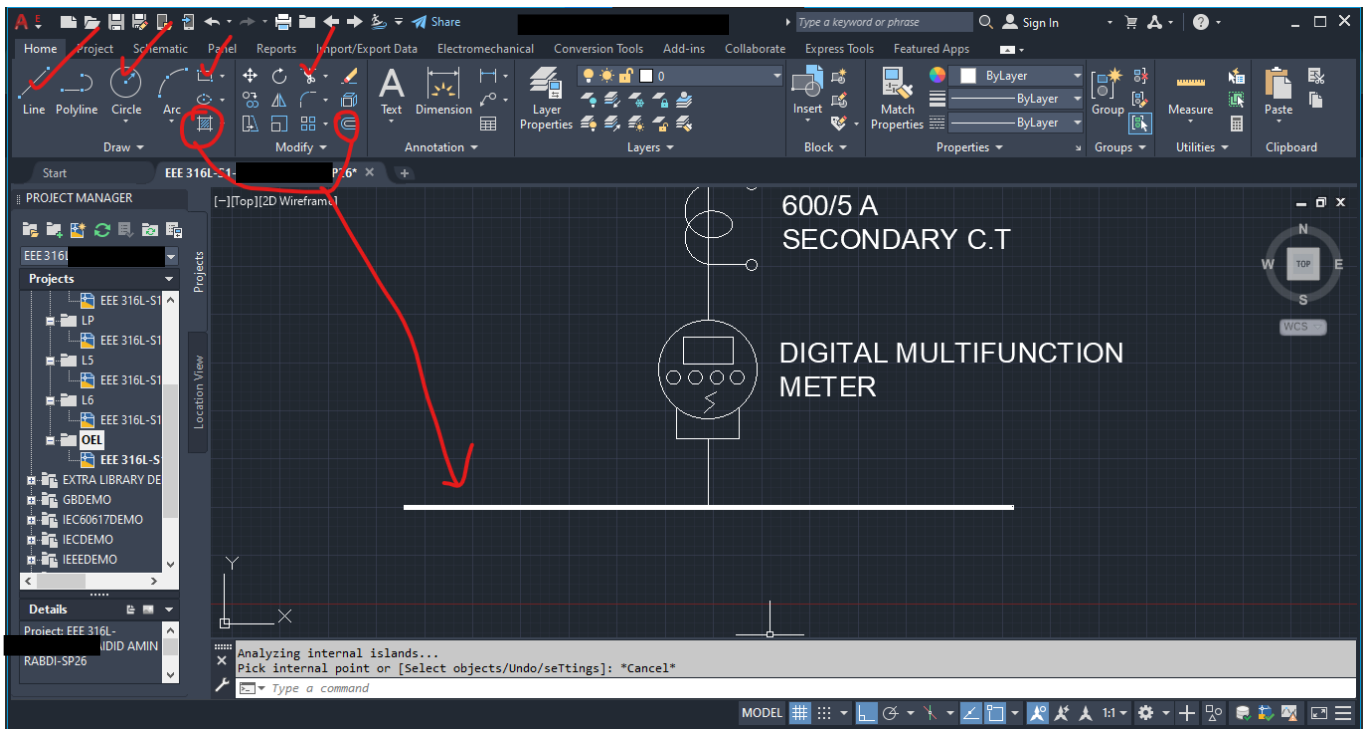
4 coil along with the line copied from the transformer design.



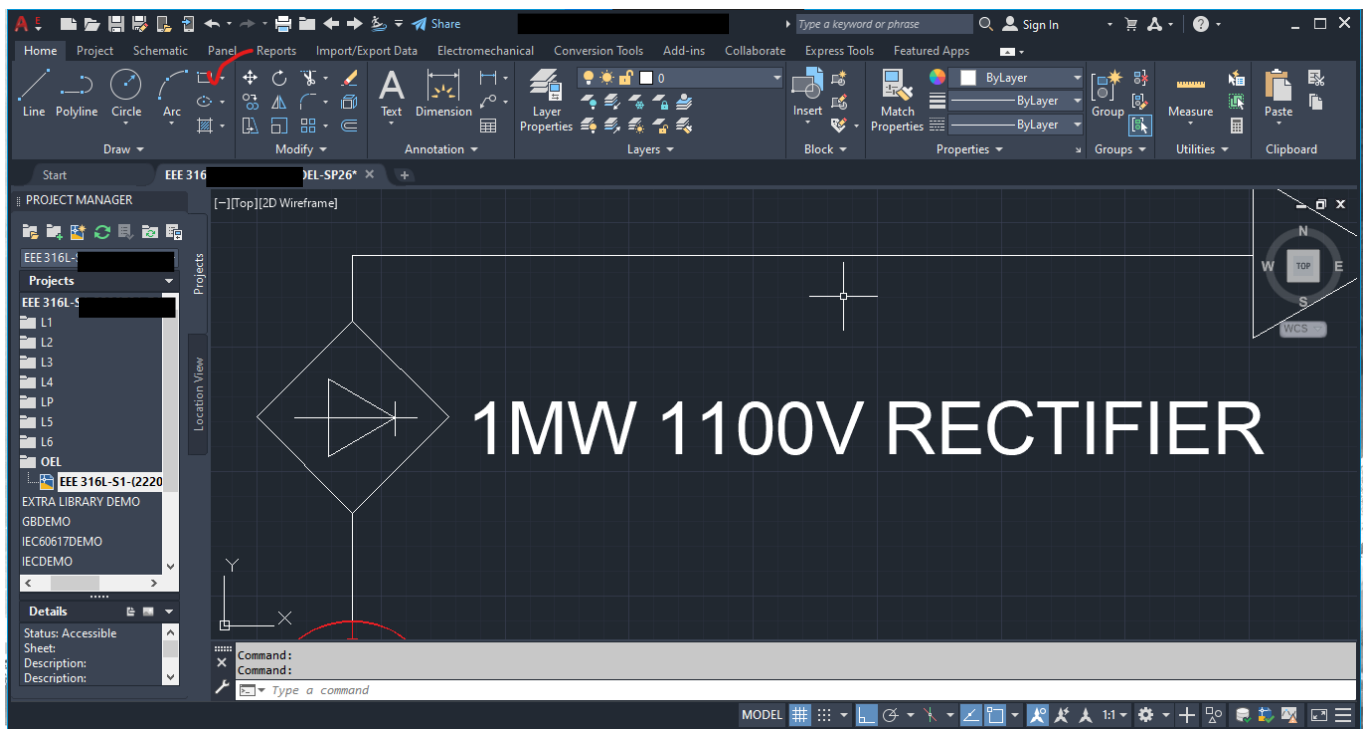
The horizontal line and an arrow designed with LINE tool bolded using OFFSET (1") tool. Then HATCH the inside. TEXT tool used to add text, SCALE tool used to make the text bigger or adjusted.



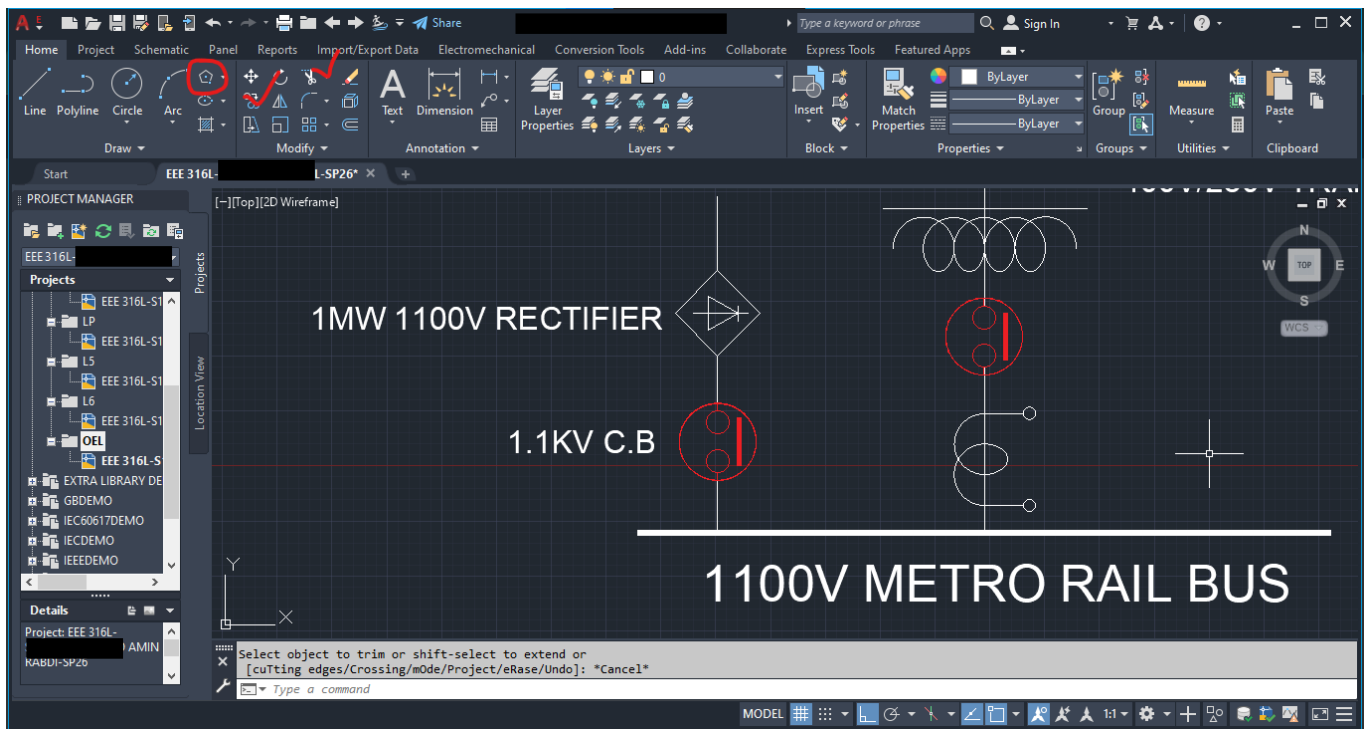
The horizontal line and an arrow designed with LINE tool bolded using OFFSET (1”) tool. Then HATCH the inside.



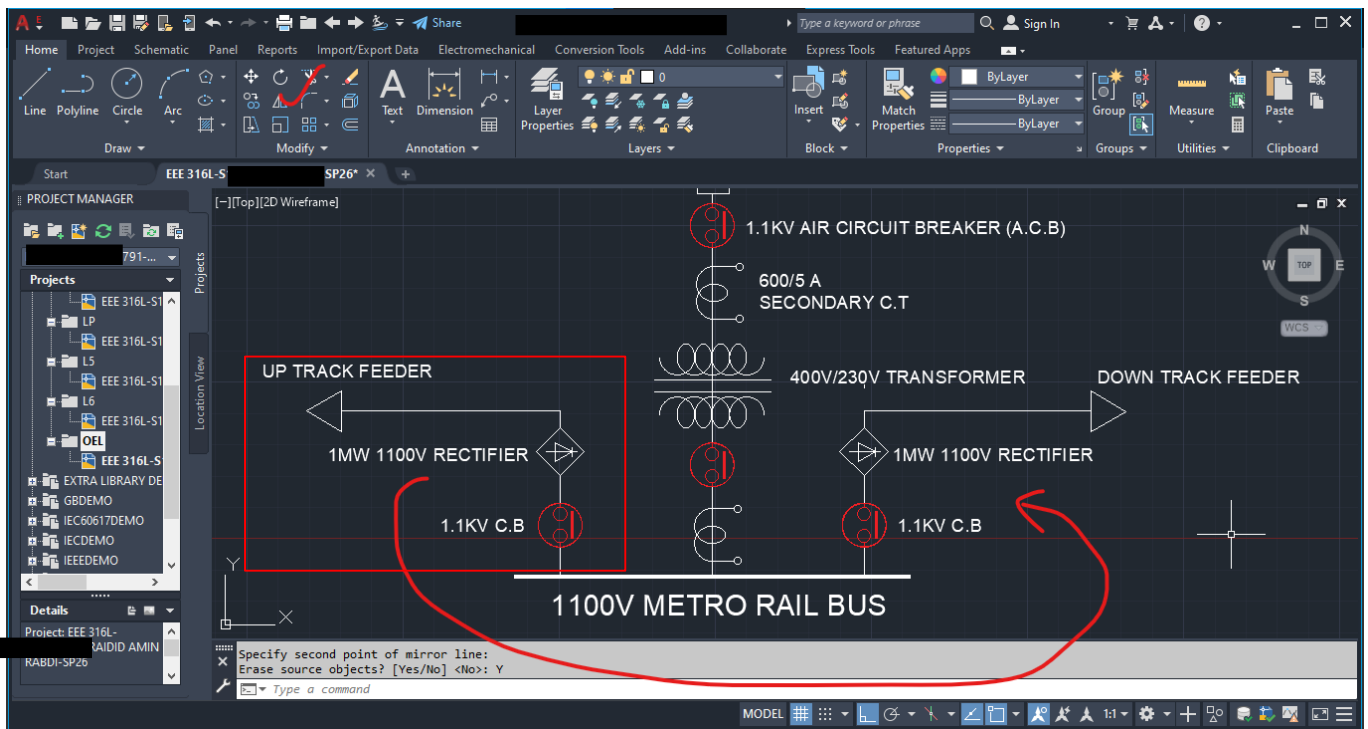
Rectangle Tool used. The triangle inside was also using the option section of Rectangle tool – POLYGON (type 3 = 3corner). Line and trim also used.



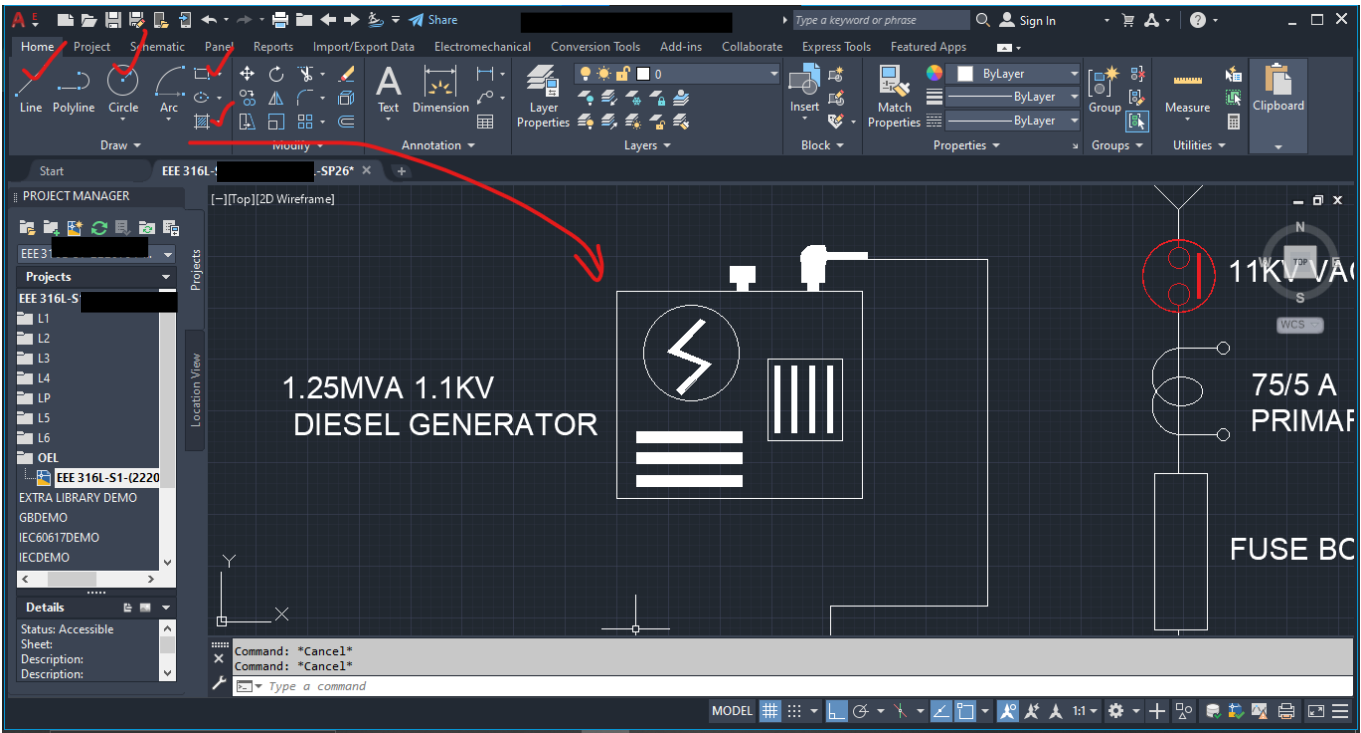
This is the design for C.B and Rectifier placed at the bottom rail bus.



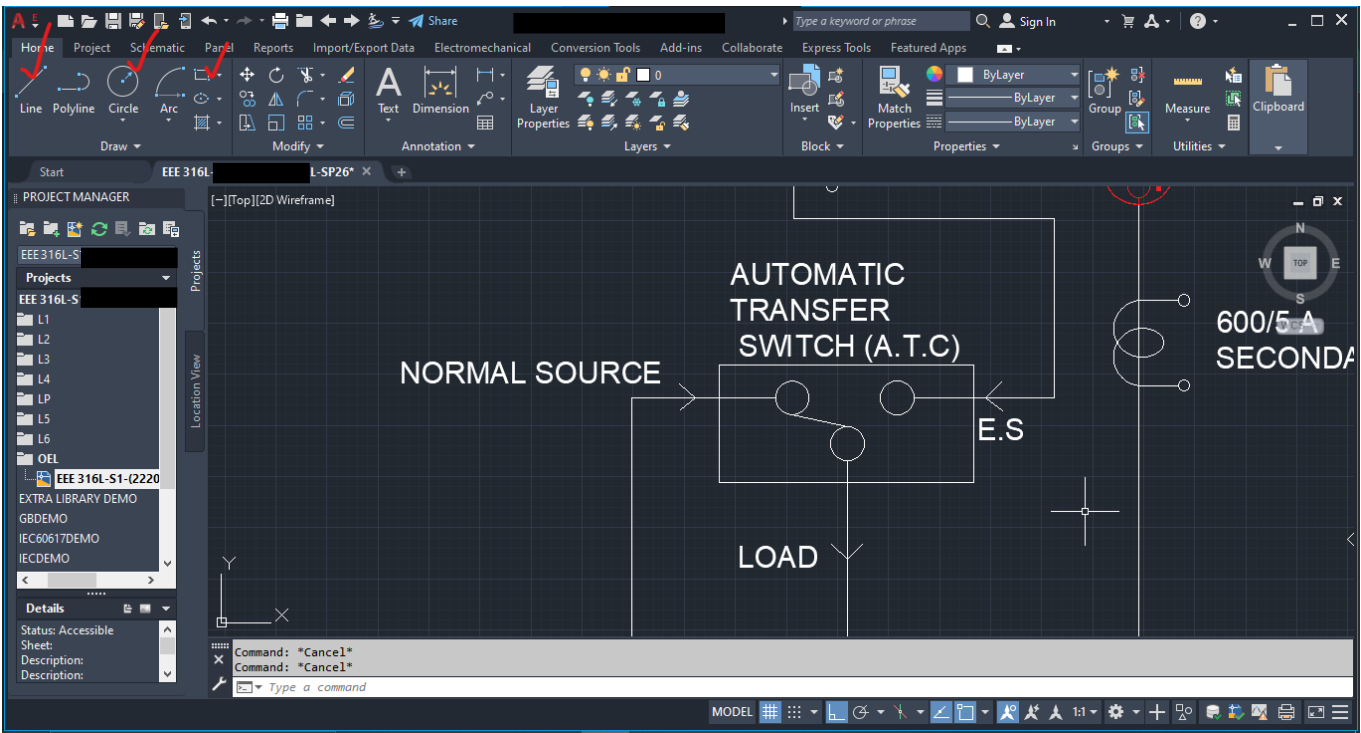
POLYGON (type 3 = 3corner) for the arrow that pointing the feeder direction. MIRROR the portion inside the box on the other side.



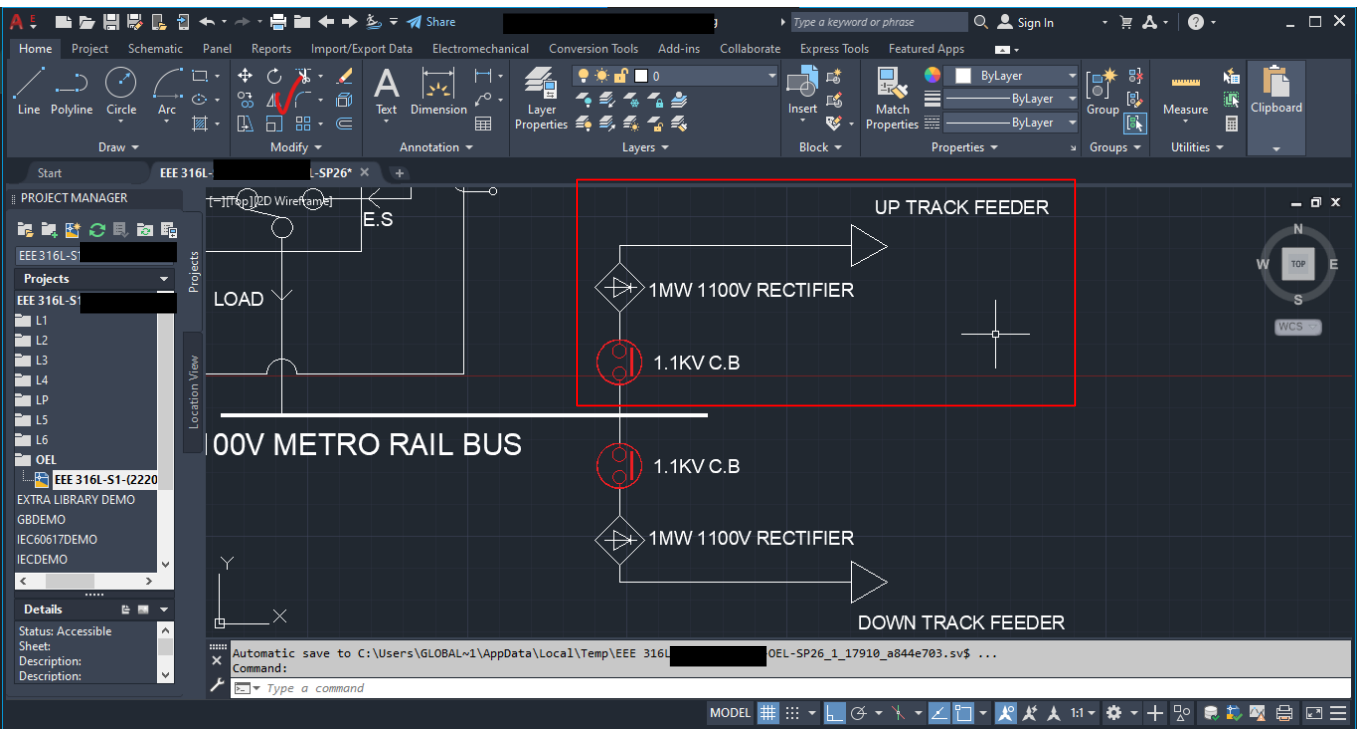
Rectangle, Line, Circle, Hatch used for this component design.



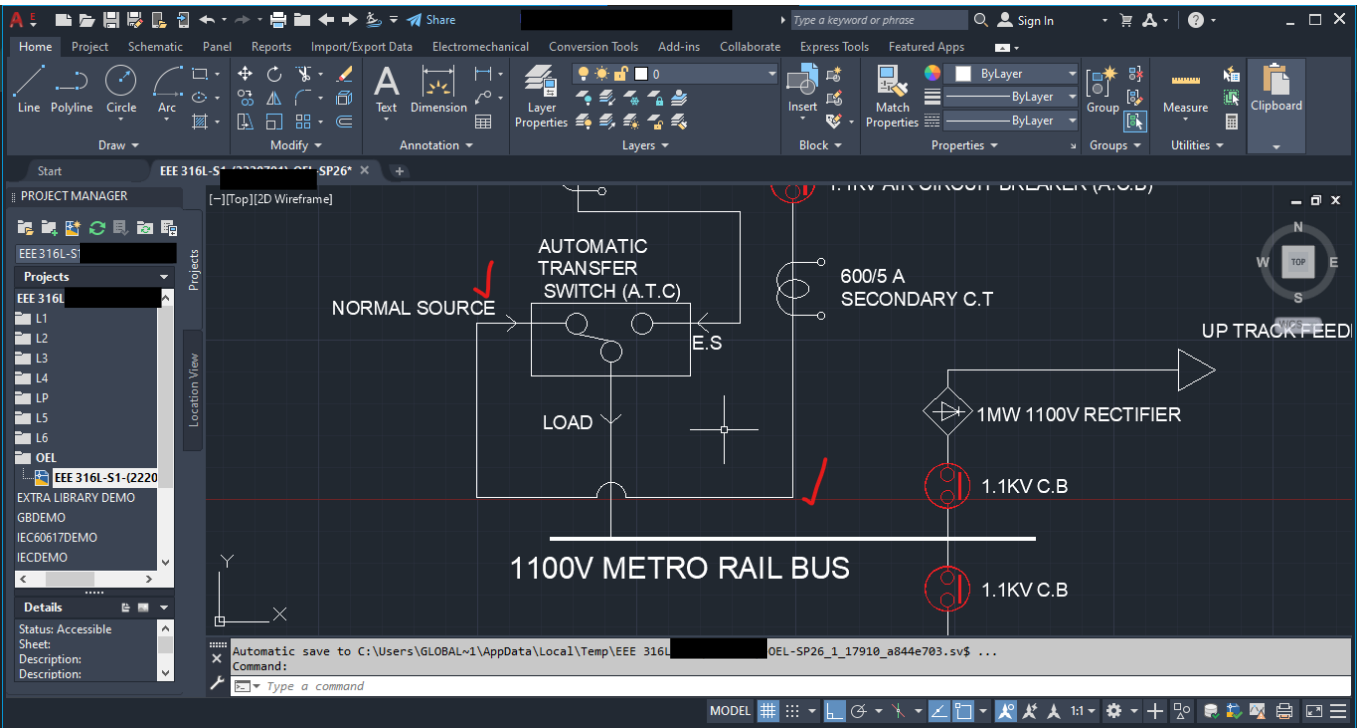
Rectangle, Line, Circle used for this component design.



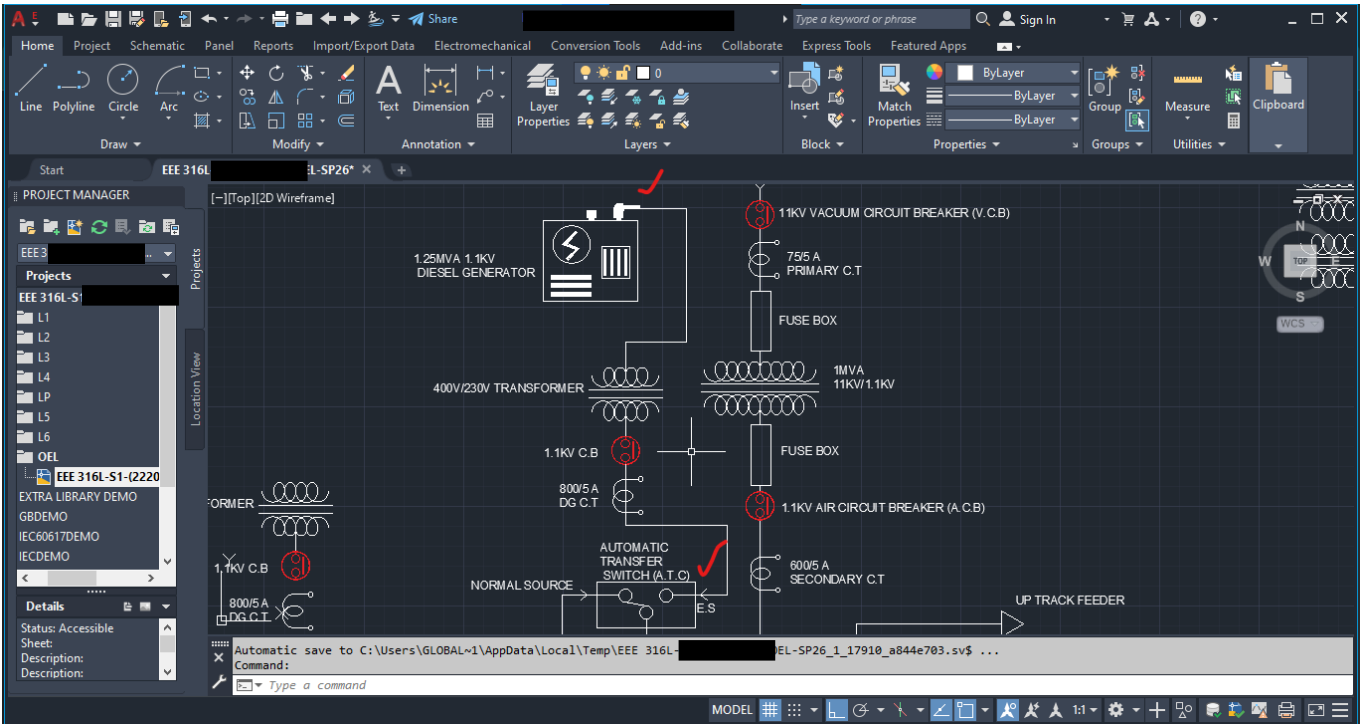
Changed the feeder placement, selected the red box portion and mirrored it downward.



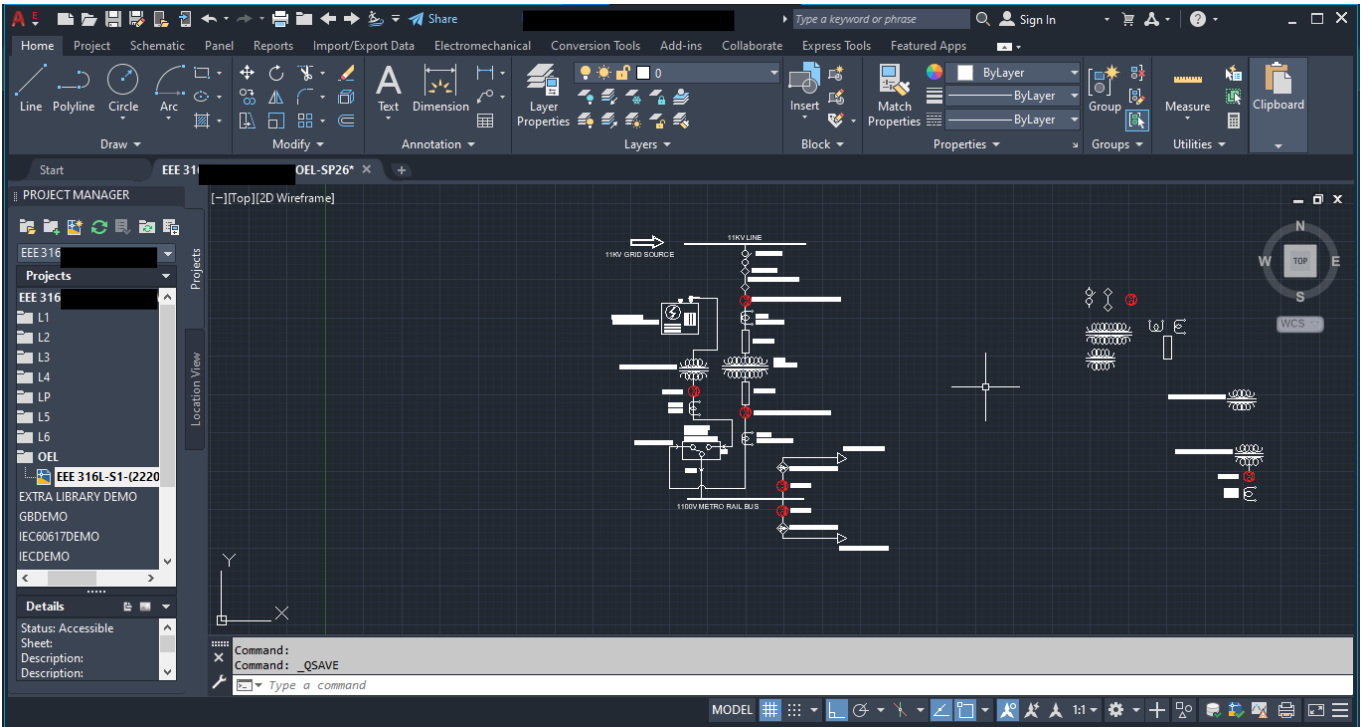
Line tool for wiring.



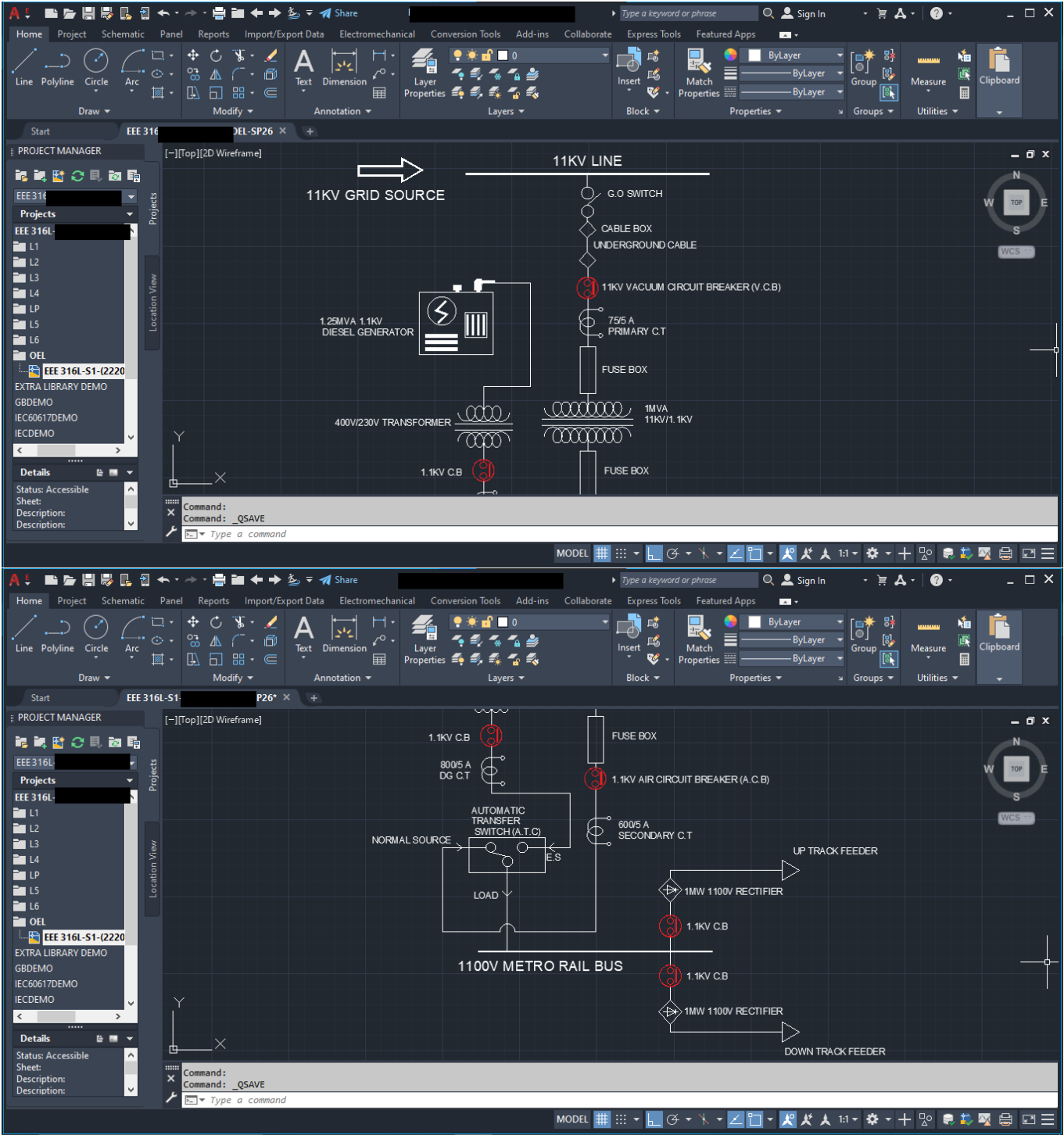
This Is the emergency line flow.



My FULL design of OPEN ENDED LAB →



ZOOMED FULL DESIGN →



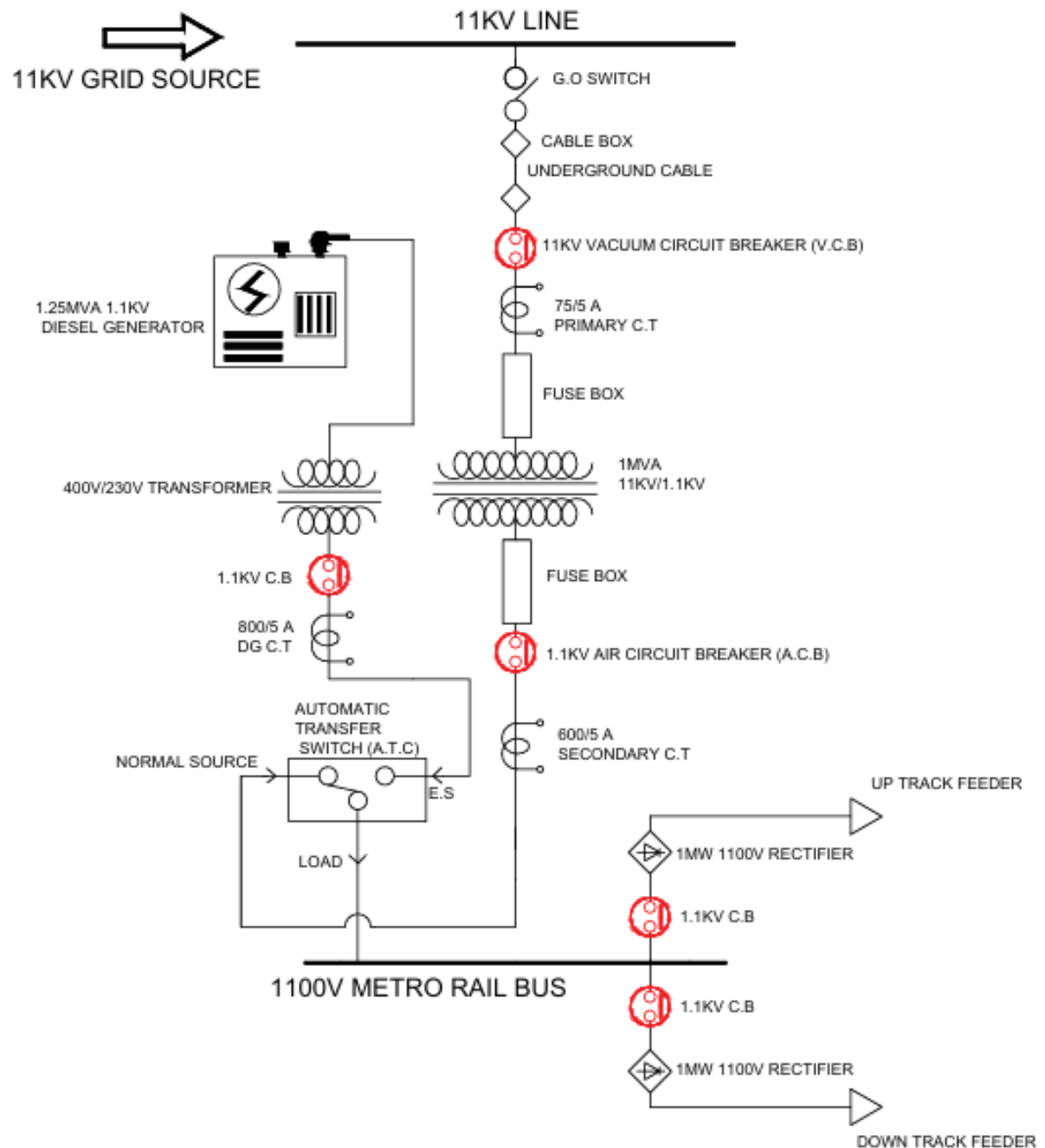


FIG: OPEN ENDED DESIGN
(Single Line Diagram of an emergency Power Station)